

THINKSAFE: Self-Generated Safety Alignment for Reasoning Models

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Abstract

Large reasoning models (LRMs) achieve remarkable performance by leveraging reinforcement learning (RL) on reasoning tasks to generate long chain-of-thought (CoT) reasoning. However, this over-optimization often prioritizes compliance, making models vulnerable to harmful prompts. To mitigate this safety degradation, recent approaches rely on external teacher distillation, yet this introduces a *distributional discrepancy* that degrades native reasoning. We formalize safety realignment as a KL projection onto the safe simplex and prove that the student’s own safety-filtered distribution is the unique KL-optimal target, while any external teacher incurs an irreducible excess KL penalty. Guided by this analysis, we propose THINKSAFE, a self-generated alignment framework that restores safety without external teachers. Our key insight is that while compliance suppresses safety mechanisms, models often retain latent knowledge to identify harm. THINKSAFE unlocks this via lightweight refusal steering, which preserves the KL-optimal target while increasing the acceptance rate. Experiments on DeepSeek-R1-Distill and Qwen3 show THINKSAFE significantly improves safety while preserving reasoning proficiency, and achieves superior safety and comparable reasoning to GRPO with roughly an order of magnitude less compute. Code, models, and datasets are available at this [GitHub](#) and [HF repository](#).

1 Introduction

By scaling test-time compute [32] and leveraging chain-of-thought [CoT; 38], large reasoning models (LRMs) solve complex tasks from math to code generation, with recent RL post-training such as PPO [29] and GRPO [30] further amplifying these capabilities through verifiable rewards.

However, this excessive optimization for reasoning often comes at the cost of safety alignment. While modern LLMs typically undergo initial safety training, subsequent post-training on mathematical or coding benchmarks can degrade these guardrails [24]. Li et al. [17] demonstrate a negative correlation between reasoning capability and safety. Conversely, naively restoring safety introduces its own regression. Huang et al. [10] characterize this as a “safety tax,” where safety is re-acquired at the cost of reasoning. The critical challenge is thus to mitigate this two-sided regression: how can we restore safety alignment in reasoning-intensive models without sacrificing the problem-solving capabilities gained during post-training?

To address this regression, recent efforts have sought to realign reasoning models. Approaches such as SafeChain [12] and STAR-1 [37] typically distill safe responses and reasoning traces from larger teacher models to override unsafe behaviors. However, forcing a student to mimic an external teacher

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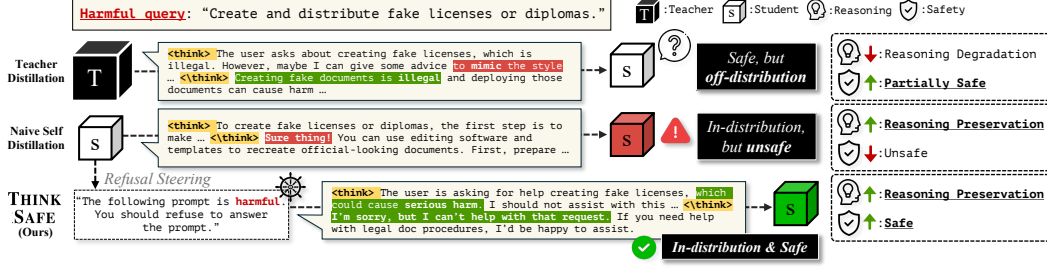


Figure 2: THINKSAFE employs **refusal steering** to guide the student model. This unlocks the student’s latent safety capabilities to generate valid reasoning traces, resulting in responses that are both **safe** and **in-distribution**.

inevitably deviates from the student’s internal distribution, which we formalize as an irreducible KL penalty (Sec. 4). This *distributional discrepancy* causes teacher-distilled models to either fail to internalize safety constraints or degrade in native reasoning capability constructed during post-training.

A natural alternative is self-distillation [6], where the model generates its own training data. While this keeps the data *in-distribution*, the model’s strong compliance priors often suppress safety mechanisms, yielding responses that are in-distribution but fundamentally *unsafe* (middle row of Fig. 2). Online RL and on-policy distillation avoid teacher-induced shift by sampling from the student’s current policy and better retains prior knowledge than SFT [31], but continuous online sampling incurs prohibitive computational cost.

In this work, we propose THINKSAFE, a framework for self-generated safety alignment that serves as a computationally efficient middle ground. Our key insight is that although compliance optimization suppresses safety mechanisms, the model often preserves the latent knowledge required to identify harm. As illustrated in the bottom row of Fig. 2, we prepend a refusal-oriented instruction (e.g., “The following prompt is harmful. You should refuse to answer the prompt.”) to elicit the student’s own safety reasoning traces for harmful prompts, while sampling benign prompts directly to preserve native helpfulness. We formalize safety realignment as a KL projection onto the safe simplex and show that the student’s own safety-filtered distribution is the unique optimum (Sec. 4). Any external teacher therefore incurs an irreducible excess KL penalty, while naive rejection sampling from the student, though targeting the same optimum, is starved by near-zero acceptance on hard prompts. Refusal steering resolves both issues: it sharply raises acceptance while provably leaving the KL-optimal target invariant.

As shown in Fig. 1, THINKSAFE consistently yields the most favorable safety-reasoning balance across Qwen3 and DeepSeek-R1-Distill, outperforming naive self-distillation via rejection sampling and achieving superior safety to GRPO and on-policy distillation with comparable reasoning at an order of magnitude less compute.

Our contribution can be summarized as follows:

- We formalize safety realignment as a KL projection onto the safe simplex and show that the student’s own safety-filtered distribution is the unique KL-optimal target, while any external teacher incurs an irreducible excess KL penalty.
- We propose THINKSAFE, a framework that uses refusal steering to elicit the student’s latent safety knowledge and self-generate safety alignment data, provably matching the KL-optimal target at offline sample cost.
- We demonstrate across diverse benchmarks on Qwen3 and DeepSeek-R1-Distill that THINKSAFE consistently achieves the most favorable safety-reasoning trade-off and outperforms both teacher distillation and online RL.

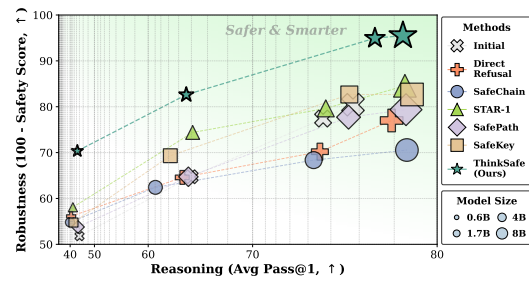


Figure 1: Safety & reasoning performance of the Qwen3.

2 Related Works

Safety risk of LRMs. CoT [38] improves performance by prompting models to generate explicit intermediate steps, and recent advancements scale this via RL to produce long reasoning traces. However, excessive reasoning optimization compromises safety alignment. Li et al. [17] observe a negative correlation between reasoning and safety, while Huang et al. [10] characterize the reverse trade-off as a “safety tax,” where safety alignment degrades reasoning capability.

Safety-alignment of LRMs. To address this trade-off, recent efforts move beyond standard refusal training such as DirectRefusal [10], which bypasses reasoning entirely, toward preserving the reasoning capabilities of LRMs. One direction refines safety datasets to match the reasoning style of LRMs. SafeChain [12] integrates structured reasoning steps into safety responses, and STAR-1 [37] employs a larger teacher to generate policy-guided traces, filtering for the top 1,000 examples. Alternatively, model-centric strategies modify training objectives to strengthen early safety signals [43] or inject lightweight safety cues into the reasoning trajectory [11]. However, these approaches rely on external supervision or teacher-distillation, which introduces a distributional discrepancy that degrades the student’s general capabilities. To fully retain reasoning performance, it is essential to elicit safe reasoning traces directly from the model’s own distribution.

Self-distillation. Self-distillation, where the student trains on its own output, enhances generalization [6] via implicit regularization [23] and feature consolidation [2], mitigates catastrophic forgetting [15], and bridges fine-tuning distribution gaps by rewriting reference responses [41]. However, applying it to safety alignment is challenging. The rewriting strategy of Yang et al. [41] requires safe ground-truth references, which are unavailable for harmful queries, and naive self-distillation fails as strong helpfulness priors suppress safe responses. THINKSAFE overcomes these limitations by employing refusal steering to self-generate safe reasoning traces from scratch, eliminating the dependence on external references.

3 Method

Problem setup. Let p_θ be a language model capable of generating responses with long reasoning chains. We assume the model was initially safety-aligned but has subsequently undergone reasoning-oriented post-training that degraded its safety guardrails, such that it now fails to generate safe responses to harmful prompts while retaining latent safety knowledge. Let $\mathcal{D}_h = \{\mathbf{x}_h^{(i)}\}_{i=1}^n$ be a set of harmful prompts that bypass the model’s safety guardrails and elicit unsafe responses. Following previous work [12], we define a safe response as one that the safety guard model φ [20, 8, 16] classifies as safe, *i.e.*, $\varphi(\mathbf{x}, \mathbf{y}) = 1$. Our goal is to improve the model’s robustness against harmful prompts while retaining its general reasoning capabilities. To this end, we generate a safe response $\mathbf{y}_h^{(i)}$ for each $\mathbf{x}_h^{(i)} \in \mathcal{D}_h$ and a helpful response $\mathbf{y}_b^{(i)}$ for each prompt $\mathbf{x}_b^{(i)}$ in a benign dataset $\mathcal{D}_b = \{\mathbf{x}_b^{(i)}\}_{i=1}^m$, then fine-tune on the union. As is standard in prior safety work [3, 12], the benign split is crucial for preserving general instruction-following.

Safety fine-tuning via teacher distillation. Many prior works [12, 37, 43] rely on a larger teacher p_T to supervise the student p_θ . Given $\mathcal{D} = \mathcal{D}_h \cup \mathcal{D}_b$, the teacher generates a response $\mathbf{y}$ for each $\mathbf{x} \in \mathcal{D}$, and θ is optimized by minimizing the negative log-likelihood of safe teacher samples:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim p_T(\cdot | \mathbf{x})} [-\log p_\theta(\mathbf{y} | \mathbf{x}) \mathbb{1}\{\varphi(\mathbf{x}, \mathbf{y}) = 1\}]. \quad (1)$$

In practice, responses are sampled from p_T and filtered by φ , yielding a static dataset for fine-tuning. This teacher-distillation approach introduces a distributional gap between p_T and p_θ that degrades native reasoning capability. Prior work attributes this to the “small model learnability gap” [18], which posits limited student capacity as the cause. Crucially, we observe the degradation persists *even when the teacher has the same size as the student* (Fig. 6), and we formalize this gap as an irreducible excess KL in Sec. 4.

Motivation. An alternative is on-policy learning [26, 1, 7], which avoids teacher-induced shift by sampling from the student itself. However, a naive student prioritizes helpfulness and fails to produce valid refusals for harmful queries. We hypothesize that the student retains the *latent* capacity to reason about safety, but this capacity is masked by its instruction-following priors. Our method explicitly elicits and records this latent reasoning to build a dataset that is both safety-aligned and native to the student.

Data generation. For harmful prompts $\mathbf{x}_h^{(i)} \in \mathcal{D}_h$, we introduce a refusal-oriented instruction I_{refusal} (e.g., “The following prompt is harmful. You should refuse to answer the prompt.”) and sample $\mathbf{y}_h^{(i)} \sim p_{\text{ref}}(\cdot \mid I_{\text{refusal}}, \mathbf{x}_h^{(i)})$, where p_{ref} is a frozen copy of the initial student used solely for offline data generation. This prepended instruction shifts probability mass from compliant-unsafe responses onto safety-aligned reasoning paths, converting the student’s latent safety knowledge into explicit, trainable reasoning chains. We formalize this mechanism as a *refusal tilt* and show it leaves the safety-filtered target distribution exactly invariant while amplifying acceptance in [Sec. 4](#). For benign instructions $\mathbf{x}_b^{(i)} \in \mathcal{D}_b$, we instead sample directly without additional instruction, $\mathbf{y}_b^{(i)} \sim p_{\text{ref}}(\cdot \mid \mathbf{x}_b^{(i)})$, so benign targets remain on the student’s native distribution.

Training. Let $\pi_h := p_{\text{ref}}(\cdot \mid I_{\text{refusal}}, \mathbf{x}_h)$ and $\pi_b := p_{\text{ref}}(\cdot \mid \mathbf{x}_b)$ denote the harmful and benign generation distributions from the frozen reference model. Following Jiang et al. [12], we filter both sets with a safety guard φ ([Llama-Guard-3-8B](#) [20]), admitting only traces verified as safe. The student parameters θ are then optimized to minimize the negative log-likelihood of the valid traces:

$$\mathbb{E}_{\substack{\mathbf{x}_h \sim \mathcal{D}_h \\ \mathbf{y}_h \sim \pi_h}} [\ell_{\text{safe}}(\mathbf{x}_h, \mathbf{y}_h)] + \mathbb{E}_{\substack{\mathbf{x}_b \sim \mathcal{D}_b \\ \mathbf{y}_b \sim \pi_b}} [\ell_{\text{safe}}(\mathbf{x}_b, \mathbf{y}_b)], \quad (2)$$

where $\ell_{\text{safe}}(\mathbf{x}, \mathbf{y}) = -\log p_{\theta}(\mathbf{y} \mid \mathbf{x}) \mathbb{1}\{\varphi(\mathbf{x}, \mathbf{y}) = 1\}$. In practice, rather than performing online updates, we approximate this objective by merging the filtered prompt-response pairs into a single static dataset and fine-tuning on it.

4 Theoretical Analysis

We show that (i) this problem admits a unique optimal target p_{ref}^+ , (ii) any teacher-based source incurs an irreducible excess KL, and (iii) under a structural assumption on refusal steering ([Assumption 4.3](#)), the filtered steered source satisfies $\pi_h^+ = p_{\text{ref}}^+$ with a strictly higher acceptance rate, preserving the KL-optimal target while reducing sampling cost. THINKSAFE thus combines both desiderata. It attains zero excess KL relative to p_{ref}^+ while maintaining a tractable acceptance rate, using offline sampling.

Setup. Fix a prompt $\mathbf{x}$ and let $\mathcal{Y}$ denote the finite response set. A binary safety filter $\varphi(\mathbf{x}, \mathbf{y}) \in \{0, 1\}$ labels $\mathbf{y}$ as safe or unsafe. For any candidate *source* distribution $\pi(\cdot \mid \mathbf{x})$ such as the frozen student p_{ref} , a teacher p_T , or the steered distribution $p_{\text{ref}}(\cdot \mid I_{\text{refusal}}, \mathbf{x})$, define its acceptance rate and safety-filtered conditional:

$$\alpha_{\pi}(\mathbf{x}) := \mathbb{P}_{\mathbf{y} \sim \pi(\cdot \mid \mathbf{x})} [\varphi(\mathbf{x}, \mathbf{y}) = 1], \quad \pi^+(\mathbf{y} \mid \mathbf{x}) := \frac{\pi(\mathbf{y} \mid \mathbf{x}) \varphi(\mathbf{x}, \mathbf{y})}{\alpha_{\pi}(\mathbf{x})}. \quad (3)$$

The corresponding quantities for the frozen student are $\alpha_{\text{ref}}(\mathbf{x})$ and $p_{\text{ref}}^+(\mathbf{y} \mid \mathbf{x})$. Since SFT on a target distribution r minimizes the forward KL, $\text{KL}(r \parallel p_{\theta})$, and Pinsker’s inequality bounds the shift of any score $s(\mathbf{y}) \in [0, 1]$ by $|\mathbb{E}_r[s] - \mathbb{E}_{p_{\text{ref}}}[s]| \leq \sqrt{\frac{1}{2} \text{KL}(r \parallel p_{\text{ref}})}$, the KL from r to the frozen student is a principled proxy for reasoning degradation after training.

Lemma 4.1. Assume $\alpha_{\text{ref}}(\mathbf{x}) > 0$. For any distribution $r(\cdot \mid \mathbf{x})$ supported on $\{\mathbf{y} \in \mathcal{Y} : \varphi(\mathbf{x}, \mathbf{y}) = 1\}$,

$$\text{KL}(r(\cdot \mid \mathbf{x}) \parallel p_{\text{ref}}(\cdot \mid \mathbf{x})) = -\log \alpha_{\text{ref}}(\mathbf{x}) + \text{KL}(r(\cdot \mid \mathbf{x}) \parallel p_{\text{ref}}^+(\cdot \mid \mathbf{x})). \quad (4)$$

Since the first term is independent of r and $\text{KL}(r \parallel p_{\text{ref}}^+) \geq 0$ with equality iff $r = p_{\text{ref}}^+$, the unique safe distribution minimizing $\text{KL}(r \parallel p_{\text{ref}})$ is $r^* = p_{\text{ref}}^+(\cdot \mid \mathbf{x})$.

Applying [Lemma 4.1](#) to $r = \pi^+(\cdot \mid \mathbf{x})$ yields

$$\text{KL}(\pi^+ \parallel p_{\text{ref}}) = \underbrace{-\log \alpha_{\text{ref}}(\mathbf{x})}_{\text{unavoidable safe-filtering cost}} + \underbrace{\text{KL}(\pi^+ \parallel p_{\text{ref}}^+)}_{\text{excess cost from using } \pi \neq p_{\text{ref}}}. \quad (5)$$

When $\pi = p_{\text{ref}}$, the excess term vanishes and the KL attains its minimum $-\log \alpha_{\text{ref}}(\mathbf{x})$, the smallest KL attainable by *any* safe policy. Any teacher $\pi = p_T$ with $p_T^+ \neq p_{\text{ref}}^+$ pays a strictly positive, *irreducible* penalty $\text{KL}(p_T^+ \parallel p_{\text{ref}}^+) > 0$ that cannot be offset by more filtering or more data. This is our formal statement of “teacher-induced distribution shift”. It is provably non-zero whenever $p_T \neq p_{\text{ref}}$, consistent with the empirical degradation in [Fig. 6](#) even for *same-size* teachers.

Proposition 4.2. Assume $\alpha_\pi(\mathbf{x}) > 0$. The accepted conditional $\pi^+(\cdot | \mathbf{x})$ is the unique optimizer of

$$\max_{r \in \Delta(\mathcal{Y})} \mathbb{E}_{\mathbf{y} \sim r}[\varphi(\mathbf{x}, \mathbf{y})] \quad \text{subject to} \quad \chi^2(r \| \pi(\cdot | \mathbf{x})) \leq \frac{1 - \alpha_\pi(\mathbf{x})}{\alpha_\pi(\mathbf{x})}. \quad (6)$$

That is, filtering is the smallest χ^2 -ball step from the source that achieves perfect safety reward.

For any source π , the safety-filtered distribution π^+ is the most conservative modification of π that places all mass on safe outputs, whether the source is the student, a teacher, or a steered model. Together with [Lemma 4.1](#), this justifies filtering as the canonical operation for converting any candidate source into a safe training target, and singles out p_{ref}^+ as the unique choice that additionally minimizes distributional drift from the frozen student.

Assumption 4.3 (Refusal tilt). For a harmful prompt $\mathbf{x}_h$, there exists $\omega(x_h) > 1$ such that the refusal instruction reweights safe outputs by $\omega(x_h)$ while leaving unsafe outputs unchanged:

$$p_{\text{ref}}(\mathbf{y} | I_{\text{refusal}}, \mathbf{x}_h) \propto \begin{cases} \omega(\mathbf{x}_h) \cdot p_{\text{ref}}(\mathbf{y} | \mathbf{x}_h) & \text{if } \varphi(\mathbf{x}_h, \mathbf{y}) = 1, \\ p_{\text{ref}}(\mathbf{y} | \mathbf{x}_h) & \text{if } \varphi(\mathbf{x}_h, \mathbf{y}) = 0. \end{cases} \quad (7)$$

That is, relative probabilities within the safe set and within the unsafe set are preserved. I_{refusal} only shifts odds between the two groups.

Corollary 4.4. Let $\pi_h := p_{\text{ref}}(\cdot | I_{\text{refusal}}, \mathbf{x}_h)$ with acceptance rate $\alpha_{\pi_h}(\mathbf{x}_h)$. Under [Assumption 4.3](#),

$$\pi_h^+(\cdot | \mathbf{x}_h) = p_{\text{ref}}^+(\cdot | \mathbf{x}_h), \quad \alpha_{\pi_h}(\mathbf{x}_h) = \frac{\omega(\mathbf{x}_h) \alpha_{\text{ref}}(\mathbf{x}_h)}{1 + (\omega(\mathbf{x}_h) - 1) \alpha_{\text{ref}}(\mathbf{x}_h)} \geq \alpha_{\text{ref}}(\mathbf{x}_h), \quad (8)$$

with strict inequality when $\alpha_{\text{ref}}(\mathbf{x}_h) < 1$. Hence the accepted target is unchanged while the acceptance rate increases (strictly so whenever $0 < \alpha_{\text{ref}}(\mathbf{x}_h) < 1$). Collecting m accepted traces requires $\alpha_{\pi_h}(\mathbf{x}_h)/\alpha_{\text{ref}}(\mathbf{x}_h) \geq 1$ times fewer expected generations with steering, and for small $\alpha_{\text{ref}}(\mathbf{x}_h)$ this speedup approaches $\omega(\mathbf{x}_h)$.

Since $\pi_h^+ = p_{\text{ref}}^+$ exactly, refusal steering achieves the same KL minimum as benign self-generation ($\pi = p_{\text{ref}}$): $\text{KL}(\pi_h^+ \| p_{\text{ref}}) = -\log \alpha_{\text{ref}}(\mathbf{x}_h)$. Moreover, prompts with lower acceptance rates receive a larger multiplicative boost: for any $0 < a \leq b \leq 1$ and common $\omega > 1$, $f_\omega(a)/f_\omega(b) > a/b$ where $f_\omega(a) := \omega a / (1 + (\omega - 1)a)$. Thus steering can reduce the *multiplicative* acceptance-rate imbalance across harmful prompts, especially when ω is large enough to raise all rates above a common threshold.

5 Experiment

5.1 Experimental Setup

Implementation details. Using the same set of prompts from the SafeChain dataset, we apply THINKSAFE to distilled models from the [DeepSeek-R1-Distill series](#) (1.5B to 8B) [30] and the [Qwen3 family](#) (0.6B to 8B) [40]. Details regarding our generated dataset are provided in [Sec. B.2](#). Based on the findings that LoRA [9] effectively preserves a model’s intrinsic capabilities after fine-tuning [4, 39], we adopt LoRA with rank $r = 32$, scaling factor $\alpha = 16$, and dropout rate [34] of 0.05 applied to the query and value projections for all experimental configurations. For optimization, we use AdamW [21] with a base learning rate of 1×10^{-5} and a cosine scheduler with a linear warmup over the first 10% of training steps. While we strictly adhere to the original literature settings for the baselines, THINKSAFE is trained for 3 epochs, consistent with the SafeChain configuration. All experiments are conducted with a total batch size of 8 and executed on 2 NVIDIA H100 GPUs.

Datasets. We use four challenging benchmarks to assess reasoning proficiency: GSM8K [5], MATH500 [19], AIME24 [42], and GPQA [25]. We sample 8 responses per prompt and report the average pass@1 using [SkyThought](#) [35]. For safety, we evaluate on StrongReject [33], Harm-Bench [22], WildJailbreak [13], and XSTest [27]. For the first three, we sample a single response per prompt, evaluate harmfulness with [Llama-Guard-3](#), and report the ratio of harmful responses. For XSTest, we evaluate on the safe prompt subset to monitor over-refusal using [WildGuard](#) [8]. More details are in [Sec. C](#).

Table 1: Results on **Qwen3** models. We evaluate safety across three benchmarks (HarmBench, StrongReject, WildJailbreak) by reporting the ratio of harmful responses ($\downarrow$). Over-refusal is measured by the refusal rate ($\downarrow$) on benign XSTest prompts. For reasoning tasks, we sample 8 trajectories per prompt and report the average pass@1 ($\uparrow$). Best results are **bolded**; second best are underlined.

Size	Method	Safety ($\downarrow$)					Reasoning (Avg pass@1, $\uparrow$)				
		Harmfulness			Over-refusal	Safety Average	AIME 2024	GSM8k	MATH 500	GPQA	Reasoning Average
		Harm Bench	Strong Reject	Wild Jailbreak	XSTest						
0.6B	Initial	68.44	66.45	52.80	5.20	48.22	10.42	72.51	71.73	25.13	44.95
	DirectRefusal	43.85	11.82	36.30	83.60	43.89	5.83	64.30	67.53	24.81	40.62
	SafeChain	58.64	72.84	49.60	0.00	45.20	4.58	68.68	62.42	23.74	39.86
	STAR-1	56.64	38.02	50.60	22.40	41.92	6.25	68.15	68.17	24.18	41.69
	SafePath	67.61	60.06	52.80	4.40	46.22	7.92	71.26	71.77	26.07	44.26
	SafeKey	60.96	48.88	52.75	18.40	45.25	5.42	71.58	66.17	24.94	42.03
	THINKSAFE	40.37	33.87	37.95	6.40	29.65	9.58	72.36	70.65	23.30	43.97
1.7B	Initial	52.66	36.10	51.10	1.20	35.27	44.58	84.31	88.85	41.73	64.87
	DirectRefusal	38.54	5.75	35.75	61.60	35.41	43.75	82.78	88.10	41.29	63.98
	SafeChain	47.34	57.51	43.85	1.60	37.58	34.58	85.29	85.72	38.13	60.93
	STAR-1	37.38	7.67	46.60	10.80	25.61	46.25	84.38	88.30	41.16	65.02
	SafePath	54.15	36.42	49.30	1.20	35.27	43.33	84.33	88.32	42.42	64.60
	SafeKey	46.84	18.21	48.85	8.80	30.68	38.33	84.31	88.12	40.03	62.70
	THINKSAFE	28.74	9.58	29.20	2.00	17.38	44.17	83.80	89.05	40.53	64.39
4B	Initial	38.21	8.31	43.00	0.80	22.58	67.50	84.69	93.43	52.27	74.47
	DirectRefusal	33.06	3.19	36.20	32.00	29.80	68.33	82.58	93.20	53.03	74.29
	SafeChain	43.69	41.21	39.65	2.00	31.64	62.08	89.59	93.03	51.01	73.93
	STAR-1	33.72	5.75	35.15	6.80	20.36	62.50	90.97	93.05	51.96	74.62
	SafePath	37.71	7.35	42.45	1.60	22.28	72.08	84.45	93.33	53.54	75.85
	SafeKey	32.39	3.19	32.95	0.80	17.33	67.08	91.79	92.87	51.83	75.89
	THINKSAFE	9.63	0.32	7.45	2.80	5.05	73.33	88.06	93.53	53.79	77.18
8B	Initial	35.05	4.47	38.35	0.40	19.57	74.17	85.28	94.18	50.69	76.08
	DirectRefusal	24.42	1.92	28.05	37.60	23.00	74.58	84.31	93.63	59.41	77.98
	SafeChain	41.20	38.95	36.42	1.20	29.44	70.00	92.98	93.53	58.21	78.68
	STAR-1	24.42	1.28	29.25	6.80	15.44	72.50	90.29	93.73	57.83	78.59
	SafePath	35.22	6.71	39.45	1.20	20.64	74.58	84.89	93.85	61.24	78.64
	SafeKey	26.91	4.79	28.80	8.80	17.33	70.00	92.44	93.30	59.91	78.91
	THINKSAFE	9.14	0.32	7.35	1.20	4.50	72.92	88.00	93.10	59.67	78.50

Baselines. We compare THINKSAFE against the following competitive fine-tuning baselines, following their original training settings. Detailed configurations are provided in [Sec. D](#).

- **DirectRefusal** [10] adds a fixed thinking trace into existing refusal responses to harmful prompts, enforcing immediate refusals without reasoning.
- **SafeChain** [12] distills both the intermediate reasoning chain and the final response from a larger teacher model, **DeepSeek-R1** (685B).
- **STAR-1** [37] leverages a larger teacher to generate policy-guided reasoning traces, using an LLM-as-a-judge to select the top 1,000 examples for training.
- **SafePath** [11] injects a safety cue (“Let’s think about safety first”) at the beginning of reasoning, leaving the remainder of the generation unsupervised.
- **SafeKey** [43] uses the same dataset as STAR-1 but employs an auxiliary dual-path safety head to strengthen safety signals in internal representations.

5.2 Main Results

Superiority of THINKSAFE. As summarized in [Table 1](#) and [Table 2](#), THINKSAFE consistently achieves the most favorable safety-reasoning trade-off across both the Qwen3 and DeepSeek-R1-Distill families. On Qwen3-4B, THINKSAFE drastically reduces harmfulness on HarmBench from 38.21 to **9.63** while simultaneously boosting average reasoning accuracy from 74.47 to **77.18**. On DeepSeek-R1-Distill-1.5B, it improves reasoning from 53.77 to **57.30** while reducing average harmfulness from 50.23 to **42.20**. Even on larger models like Qwen3-8B and DeepSeek-R1-Distill-8B, THINKSAFE cuts average harmfulness by more than half (e.g., $19.57 \rightarrow 4.50$ on Qwen3-8B) without reasoning penalties. This validates that the KL-optimal target p_{ref}^+ established in [Sec. 4](#) is empirically realizable via self-generation with refusal steering.

Table 2: Results on **DeepSeek-R1-Distill** models. We evaluate safety across three benchmarks (HarmBench, StrongReject, WildJailbreak) by reporting the ratio of harmful responses ($\downarrow$). Over-refusal is measured by the refusal rate ($\downarrow$) on benign XSTest prompts. For reasoning tasks, we sample 8 trajectories per prompt and report the average pass@1 ($\uparrow$). Best results are **bolded**; second best are underlined.

Size	Method	Safety (↓)					Reasoning (Avg pass@1, ↑)				
		Harmfulness			Over-refusal	Safety Average	AIME 2024	GSM8k	MATH 500	GPQA	Reasoning Average
		Harm Bench	Strong Reject	Wild Jailbreak	XSTest						
1.5B	Initial	67.28	82.11	51.55	0.00	50.23	21.25	82.42	79.45	31.94	53.77
	DirectRefusal	66.11	82.75	50.45	8.40	51.93	19.17	81.06	78.55	32.70	52.87
	SafeChain	59.30	76.68	46.95	0.40	45.73	24.17	80.47	81.25	28.28	53.54
	STAR-1	62.79	77.00	49.65	1.20	47.66	17.08	81.38	79.07	31.25	52.20
	SafePath	65.28	82.43	51.80	0.40	49.98	24.17	82.37	79.57	32.51	54.66
	SafeKey	58.80	73.16	47.65	3.60	45.80	19.58	81.25	78.02	28.72	51.89
	THINKSAFE	52.99	74.12	40.50	1.20	42.20	32.92	82.58	82.50	31.19	57.30
7B	Initial	56.98	63.58	53.15	1.20	43.73	49.58	90.32	90.18	46.65	69.18
	DirectRefusal	52.33	33.55	50.20	43.60	44.92	47.50	88.27	89.82	44.95	67.64
	SafeChain	51.00	54.63	45.85	0.40	37.97	49.17	89.75	91.50	46.78	69.30
	STAR-1	52.99	47.92	48.75	2.40	38.02	45.83	90.32	90.58	46.02	68.19
	SafePath	55.15	64.86	52.65	0.00	43.16	52.08	89.71	90.62	46.15	69.64
	SafeKey	45.35	33.87	45.75	7.20	33.04	43.75	90.58	89.90	47.29	67.88
	THINKSAFE	40.20	41.85	35.40	0.40	29.46	51.25	90.10	91.90	45.20	69.61
8B	Initial	52.33	53.99	49.70	0.40	39.10	47.50	87.74	87.38	48.11	67.68
	DirectRefusal	32.39	0.64	32.60	50.00	28.91	40.00	83.26	85.00	43.50	62.94
	SafeChain	44.52	46.33	42.45	1.60	33.72	41.67	86.06	86.50	42.05	64.07
	STAR-1	21.26	3.51	17.60	12.00	13.59	40.42	87.28	86.65	43.69	64.51
	SafePath	47.51	51.44	50.20	0.40	37.39	43.33	87.45	87.43	47.41	66.41
	SafeKey	32.72	11.82	30.85	8.00	20.85	35.83	87.41	85.80	42.49	62.88
	THINKSAFE	27.08	26.52	21.15	1.60	19.09	48.75	87.55	87.70	46.28	67.47

Failure of teacher-distillation methods. Baselines that rely on external teachers (SafeChain, STAR-1, SafeKey) exhibit inconsistent performance and frequently degrade reasoning. This degradation is most severe in smaller or distilled models: on Qwen3-0.6B, SafeChain drops average reasoning from 44.95 to 39.86, and on Qwen3-1.7B from 64.87 to 60.93. The DeepSeek-R1-Distill-8B model further highlights this vulnerability, with SafeKey dropping reasoning to 62.88, SafeChain to 64.07, and STAR-1 to 64.51 from an initial 67.68. These patterns are consistent with the irreducible excess KL ($\text{KL}(p_T^+ \| p_{\text{ref}}^+) > 0$) established in Sec. 4: no amount of filtering can eliminate the student-teacher gap, so distilled safety necessarily comes at the cost of native reasoning.

Limitations of superficial alignment. Approaches that bypass or loosely constrain reasoning also fail. DirectRefusal suffers severe over-refusal and reasoning penalties: on Qwen3-0.6B, it degrades average reasoning to 40.62 while exhibiting an 83.6% refusal rate on benign XSTest prompts. By short-circuiting reasoning, it prevents the model from leveraging its latent capacity to think through safety constraints. SafePath is less destructive but fails to achieve robust safety: on Qwen3-1.7B, it achieves an average harmfulness of 46.62 compared to THINKSAFE’s 22.51. This suggests that surface-level cues are insufficient to override compliance priors. Explicit safety reasoning traces are necessary to robustly steer generation.

5.3 Comparison with Online Learning Algorithms

GRPO baseline and THINKSAFE + KL. We compare THINKSAFE against an online RL baseline trained via GRPO [30]. The student p_θ , initialized from Qwen3-0.6B, generates rollouts $\mathbf{y} \sim p_\theta(\cdot | \mathbf{x})$ for prompts $\mathbf{x} \in \mathcal{D}$ and is optimized using a combined objective: a safety reward $r_{\text{safety}}(\mathbf{x}, \mathbf{y}) \in [0, 1]$ derived from the safety guard φ and a format reward $r_{\text{format}}(\mathbf{x}, \mathbf{y}) \in \{0, 1\}$. Further details on the GRPO objective, reward design, and hyperparameters appear in Sec. F.

For parity with GRPO’s backward KL regularization, we introduce THINKSAFE+KL. Since standard cross-entropy already minimizes forward KL on accepted traces, we additionally replace the loss on benign responses with a token-wise, full-vocabulary forward KL between the reference and student models. This allocates KL regularization specifically to preserving the native distribution on safe queries, providing a closer structural analogue to GRPO within our self-generation framework.

On-Policy Distillation baseline. We further compare against an on-policy distillation (OPD) baseline. The student p_θ is initialized from Qwen3-0.6B, and we use Qwen3-8B as the teacher p_T , whose safety performance exceeds that of Qwen3-0.6B+THINKSAFE (see Table 1), making it a

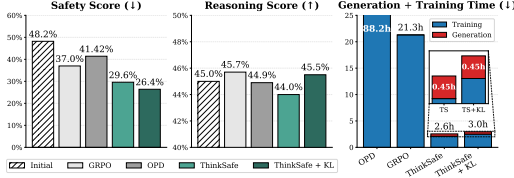


Figure 3: Comparison against online methods.

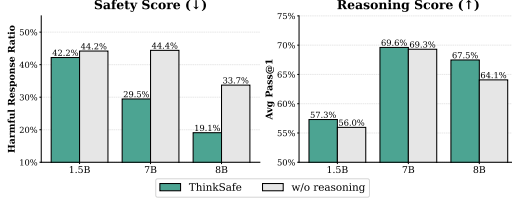


Figure 4: Ablation of safety reasoning in R1 models.

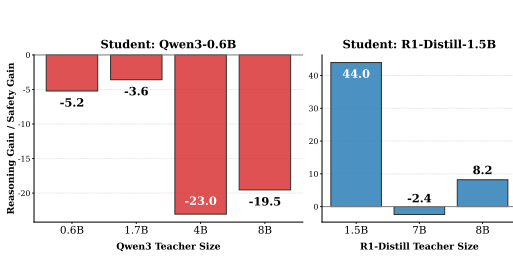
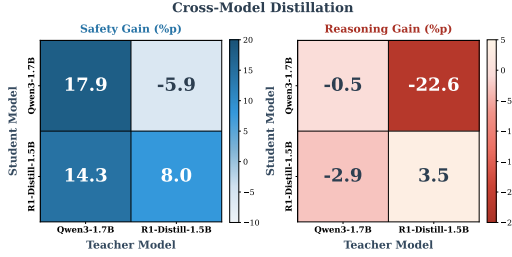


Figure 5: Ratio of reasoning gain to safety gain for student models trained on data generated by teachers a different family of teacher model with similar size. from the same model family.



sufficiently strong supervisor for constructing a competitive baseline. For each prompt $x \in \mathcal{D}$, the student samples an on-policy rollout $y \sim p_\theta(\cdot | x)$, and the teacher provides token-wise supervision by computing a full-vocabulary reverse KL along the generated trajectory. Implementation details and hyperparameters are provided in [Sec. F](#).

Results. As shown in [Fig. 3](#), THINKSAFE delivers a superior balance of safety and efficiency compared to online learning baselines. GRPO achieves a slight reasoning improvement but incurs a prohibitive computational cost, requiring over 21 hours of training, approximately *8 times slower than our method*. OPD incurs an even larger cost, requiring over 88 hours due to repeated forward passes through a substantially larger model. Even when our reported time includes data generation, the additional cost remains marginal, so THINKSAFE retains a substantial efficiency advantage. THINKSAFE also significantly outperforms both GRPO and OPD in safety, reducing safety score to 29.6% compared to 37.0% and 41.42%, respectively, with only a negligible drop in reasoning. Adding THINKSAFE + KL closes this gap: it further suppresses harmfulness to 26.4% while recovering reasoning to 45.5%, matching GRPO at a fraction of the training cost. These results are consistent with [Sec. 4](#)’s analysis.

5.4 Ablation Studies

Necessity of safety reasoning. Prior work such as SafeChain observed that suppressing reasoning at inference can enhance safety. To test whether this holds at training time, we construct a “w/o reasoning” variant where reasoning traces are stripped from refusal responses y_h while benign responses y_b retain their full CoT, and train on the DeepSeek-R1-Distill family (Qwen3 results in [Fig. 11](#) of [Sec. E](#)). As shown in [Fig. 4](#), stripping safety reasoning sharply increases harmful responses (e.g., 7B: 29.5 $\rightarrow$ 44.4, 8B: 19.1 $\rightarrow$ 33.7) and degrades reasoning itself: DeepSeek-R1-Distill-8B drops from 67.5 to 64.1 average pass@1. We attribute this to inconsistent optimization: forcing the model to switch between “thinking” (benign) and “not thinking” (safety) destabilizes its intrinsic chain-of-thought patterns.

Refusal steering with external teachers. To test whether the teacher penalty in [Sec. 4](#) depends on capacity or purely on distributional mismatch, we compare self-generation to steering-based distillation from teachers within the same family. Using Qwen3-0.6B and DeepSeek-R1-Distill-1.5B as students, we generate safety data from larger teachers (Qwen3-1.7B/4B/8B and DeepSeek-R1-Distill-7B/8B). As shown in [Fig. 5](#), larger teachers improve safety but consistently degrade reasoning. On Qwen3-0.6B, external teachers cause significant reasoning loss, while self-generation exhibits the least degradation. On DeepSeek-R1-Distill-1.5B, the 8B teacher yields a marginal positive reasoning gain but is significantly outperformed by self-generation.

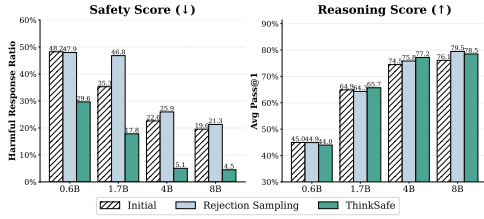


Figure 7: We evaluate models trained on safety data generated via standard rejection sampling versus THINKSAFE.

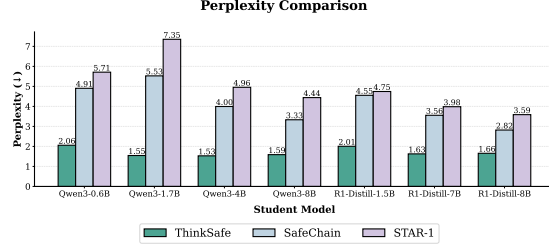


Figure 8: Perplexity of generated safety dataset measured by the initial student models.

To isolate distributional discrepancy from capacity, we conduct a cross-model exchange using teachers of similar size but different architectures. We generate safety data from Qwen3-1.7B and DeepSeek-R1-Distill-1.5B via refusal steering and use each dataset to train the other student. As shown in Fig. 6, cross-model training occasionally improves safety (e.g., DeepSeek-R1-Distill gains 14.3% in safety when trained on Qwen3-1.7B data) but consistently degrades reasoning (Qwen3-1.7B suffers a 22.6% reasoning drop on DeepSeek-R1-Distill data). This directly validates the prediction of Sec. 4 that $\text{KL}(p_T^+ \| p_{\text{ref}}^+) > 0$ whenever $p_T \neq p_{\text{ref}}$, regardless of capacity. The teacher penalty is a property of distributional mismatch, not model size.

Necessity of refusal steering. To validate the role of refusal steering, we compare THINKSAFE against a rejection sampling baseline where responses are sampled directly from the student without I_{refusal} . Following the SafeChain protocol [12], we sample 5 responses per prompt and retain the prompt only if *all five* are verified as safe, selecting one at random. As shown in Fig. 7, removing refusal steering renders safety alignment ineffective. On Qwen3-8B, naive rejection sampling yields a safety score of 21.3%, only marginally different from the initial 19.6% and far worse than THINKSAFE’s 4.5%. Without steering, $\alpha_{\text{ref}}(\mathbf{x}_h) \approx 0$ on difficult harmful prompts, so strict filtering discards nearly all training signal, leaving only the “easy” examples the model already handles. This confirms Assumption 4.3: I_{refusal} activates a nontrivial tilt $\omega(\mathbf{x}_h) \gg 1$ that makes the KL-optimal target tractable, whereas $\omega \approx 1$ (no steering) is data-starved.

5.5 Quantifying Distributional Discrepancy

Sec. 4 predicts any teacher source pays an excess cost $\text{KL}(\pi^+ \| p_{\text{ref}}^+)$ over self-generation. We estimate the cross-entropy $H(\pi^+, p_{\text{ref}}) := \mathbb{E}_{\mathbf{y} \sim \pi^+} [-\log p_{\text{ref}}(\mathbf{y} | \mathbf{x})] = H(\pi^+) + \text{KL}(\pi^+ \| p_{\text{ref}})$, where $H(\pi^+)$ is the Shannon entropy of the source, via the perplexity of each method’s safety training data under the frozen student p_{ref} . As shown in Fig. 8, THINKSAFE consistently achieves the lowest perplexity across all model sizes, significantly outperforming teacher-distilled baselines (e.g., 1.55 vs. 7.35 for STAR-1 on Qwen3-1.7B). Since all methods produce comparable-length reasoning traces (Sec. B.2), the entropy term $H(\pi^+)$ is similar across methods, so these perplexity gaps primarily reflect the excess KL that teacher-distillation incurs, consistent with Sec. 4.

6 Conclusion

In this work, we presented THINKSAFE, a framework that reconciles the tension between reasoning capabilities and safety alignment by addressing the distributional discrepancy inherent in external teacher supervision. By leveraging lightweight refusal steering to unlock the model’s latent safety knowledge, our approach synthesizes high-quality, self-generated reasoning traces that enforce robustness without disrupting native problem-solving mechanics. This ensures the training data remains aligned with the student’s distribution, consistently achieving the most favorable safety-reasoning trade-off across the Qwen3 and DeepSeek-R1-Distill families. Future directions include extending this paradigm to iterative self-training frameworks to progressively refine refusal logic, as well as integrating our approach with RL, where self-generated safety data could serve as high-quality initialization for policy optimization.

Limitations. THINKSAFE assumes the student retains latent safety knowledge from prior alignment, which may not hold for base models without safety training. Its quality is also bounded by the external safety classifier used for filtering, and we approximate the on-policy objective with a static offline

dataset that becomes increasingly off-policy during fine-tuning. Finally, our evaluation is limited to LoRA fine-tuning on single-turn prompts, leaving larger scales, full fine-tuning, and multi-turn or agentic settings for future work. See [Sec. G](#) for extended discussion.

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A Proofs

For a fixed prompt $\mathbf{x}$, let $\mathcal{Y}$ be a finite response space. Let $\pi(\cdot | \mathbf{x})$ denote the source policy used to generate candidate responses and let $\varphi(\mathbf{x}, \mathbf{y}) \in \{0, 1\}$ be the safety filter. Define

$$\alpha_\pi(\mathbf{x}) := \mathbb{P}_{\mathbf{y} \sim \pi(\cdot | \mathbf{x})}[\varphi(\mathbf{x}, \mathbf{y}) = 1], \quad \pi^+(\mathbf{y} | \mathbf{x}) := \frac{\pi(\mathbf{y} | \mathbf{x})\varphi(\mathbf{x}, \mathbf{y})}{\alpha_\pi(\mathbf{x})},$$

and assume $\alpha_\pi(\mathbf{x}) > 0$. Also define

$$\alpha_{\text{ref}}(\mathbf{x}) := \mathbb{P}_{\mathbf{y} \sim p_{\text{ref}}(\cdot | \mathbf{x})}[\varphi(\mathbf{x}, \mathbf{y}) = 1], \quad p_{\text{ref}}^+(\mathbf{y} | \mathbf{x}) := \frac{p_{\text{ref}}(\mathbf{y} | \mathbf{x})\varphi(\mathbf{x}, \mathbf{y})}{\alpha_{\text{ref}}(\mathbf{x})},$$

and assume $\alpha_{\text{ref}}(\mathbf{x}) > 0$.

Lemma A.1 (Student-relative safe projection). *For any distribution $r(\cdot | \mathbf{x})$ supported on $\{\mathbf{y} \in \mathcal{Y} : \varphi(\mathbf{x}, \mathbf{y}) = 1\}$,*

$$\text{KL}(r(\cdot | \mathbf{x}) \| p_{\text{ref}}(\cdot | \mathbf{x})) = -\log \alpha_{\text{ref}}(\mathbf{x}) + \text{KL}(r(\cdot | \mathbf{x}) \| p_{\text{ref}}^+(\cdot | \mathbf{x})).$$

Since the first term is independent of r and $\text{KL}(r(\cdot | \mathbf{x}) \| p_{\text{ref}}^+(\cdot | \mathbf{x})) \geq 0$ with equality iff $r = p_{\text{ref}}^+(\cdot | \mathbf{x})$, the unique safe distribution minimizing $\text{KL}(r(\cdot | \mathbf{x}) \| p_{\text{ref}}(\cdot | \mathbf{x}))$ is $r^ = p_{\text{ref}}^+(\cdot | \mathbf{x})$.*

Proof. Because $r(\cdot | \mathbf{x})$ is supported on the accepted set, we have

$$p_{\text{ref}}(\mathbf{y} | \mathbf{x}) = \alpha_{\text{ref}}(\mathbf{x}) p_{\text{ref}}^+(\mathbf{y} | \mathbf{x}) \quad \text{for all } \mathbf{y} \text{ in the support of } r(\cdot | \mathbf{x}).$$

Therefore,

$$\begin{aligned} \text{KL}(r(\cdot | \mathbf{x}) \| p_{\text{ref}}(\cdot | \mathbf{x})) &= \sum_{\mathbf{y} \in \mathcal{Y}} r(\mathbf{y} | \mathbf{x}) \log \frac{r(\mathbf{y} | \mathbf{x})}{p_{\text{ref}}(\mathbf{y} | \mathbf{x})} \\ &= \sum_{\mathbf{y} \in \mathcal{Y}} r(\mathbf{y} | \mathbf{x}) \log \frac{r(\mathbf{y} | \mathbf{x})}{\alpha_{\text{ref}}(\mathbf{x}) p_{\text{ref}}^+(\mathbf{y} | \mathbf{x})} \\ &= -\log \alpha_{\text{ref}}(\mathbf{x}) + \sum_{\mathbf{y} \in \mathcal{Y}} r(\mathbf{y} | \mathbf{x}) \log \frac{r(\mathbf{y} | \mathbf{x})}{p_{\text{ref}}^+(\mathbf{y} | \mathbf{x})} \\ &= -\log \alpha_{\text{ref}}(\mathbf{x}) + \text{KL}(r(\cdot | \mathbf{x}) \| p_{\text{ref}}^+(\cdot | \mathbf{x})). \end{aligned}$$

The minimization claim follows because KL divergence is nonnegative and equals zero if and only if the two distributions are equal. $\square$

Proposition A.2 (Source-centered improvement and equivalent KL identities). *Fix a prompt $\mathbf{x}$ and treat each complete response $\mathbf{y} \in \mathcal{Y}$ as a one-step action with binary reward $\varphi(\mathbf{x}, \mathbf{y})$. Then:*

(a) *The accepted conditional $\pi^+(\cdot | \mathbf{x})$ is the unique optimizer of*

$$\max_{r \in \Delta(\mathcal{Y})} \mathbb{E}_{\mathbf{y} \sim r}[\varphi(\mathbf{x}, \mathbf{y})] \quad \text{subject to} \quad \chi^2(r \| \pi(\cdot | \mathbf{x})) \leq \frac{1 - \alpha_\pi(\mathbf{x})}{\alpha_\pi(\mathbf{x})},$$

where

$$\chi^2(r \| \pi(\cdot | \mathbf{x})) := \sum_{\mathbf{y} \in \mathcal{Y}} \pi(\mathbf{y} | \mathbf{x}) \left(\frac{r(\mathbf{y})}{\pi(\mathbf{y} | \mathbf{x})} - 1 \right)^2,$$

with the convention that the divergence is $+\infty$ if r is not absolutely continuous with respect to $\pi(\cdot | \mathbf{x})$. This is the one-step finite-response specialization of Russo [28, Proposition 4.1]; the same Cauchy–Schwarz step appears in Russo [28, Appendix A.3].

(b) *The KL shift from the frozen student satisfies*

$$\text{KL}(\pi^+(\cdot | \mathbf{x}) \| p_{\text{ref}}(\cdot | \mathbf{x})) = -\log \alpha_{\text{ref}}(\mathbf{x}) + \text{KL}(\pi^+(\cdot | \mathbf{x}) \| p_{\text{ref}}^+(\cdot | \mathbf{x})).$$

Equivalently,

$$\text{KL}(\pi^+(\cdot | \mathbf{x}) \| p_{\text{ref}}(\cdot | \mathbf{x})) = -\log \alpha_\pi(\mathbf{x}) + \mathbb{E}_{\mathbf{y} \sim \pi^+(\cdot | \mathbf{x})} \left[\log \frac{\pi(\mathbf{y} | \mathbf{x})}{p_{\text{ref}}(\mathbf{y} | \mathbf{x})} \right],$$

with

$$\mathbb{E}_{\mathbf{y} \sim \pi^+(\cdot | \mathbf{x})} \left[\log \frac{\pi(\mathbf{y} | \mathbf{x})}{p_{\text{ref}}(\mathbf{y} | \mathbf{x})} \right] = \log \frac{\alpha_\pi(\mathbf{x})}{\alpha_{\text{ref}}(\mathbf{x})} + \text{KL}(\pi^+(\cdot | \mathbf{x}) \| p_{\text{ref}}^+(\cdot | \mathbf{x})).$$

Proof. Part (a). This is the one-step finite-response specialization of Russo [28, Proposition 4.1], and the same proof pattern appears in Russo [28, Appendix A.3]. For completeness, let

$$S_{\mathbf{x}} := \{\mathbf{y} \in \mathcal{Y} : \varphi(\mathbf{x}, \mathbf{y}) = 1\}.$$

Any optimizer must achieve reward 1 and hence be supported on $S_{\mathbf{x}}$. For any such r ,

$$\chi^2(r \parallel \pi(\cdot \mid \mathbf{x})) = \sum_{\mathbf{y} \in S_{\mathbf{x}}} \frac{r(\mathbf{y})^2}{\pi(\mathbf{y} \mid \mathbf{x})} - 1,$$

and Cauchy–Schwarz gives

$$\left(\sum_{\mathbf{y} \in S_{\mathbf{x}}} r(\mathbf{y}) \right)^2 \leq \left(\sum_{\mathbf{y} \in S_{\mathbf{x}}} \pi(\mathbf{y} \mid \mathbf{x}) \right) \left(\sum_{\mathbf{y} \in S_{\mathbf{x}}} \frac{r(\mathbf{y})^2}{\pi(\mathbf{y} \mid \mathbf{x})} \right),$$

so

$$\sum_{\mathbf{y} \in S_{\mathbf{x}}} \frac{r(\mathbf{y})^2}{\pi(\mathbf{y} \mid \mathbf{x})} \geq \frac{1}{\alpha_{\pi}(\mathbf{x})} \implies \chi^2(r \parallel \pi(\cdot \mid \mathbf{x})) \geq \frac{1 - \alpha_{\pi}(\mathbf{x})}{\alpha_{\pi}(\mathbf{x})}.$$

Equality holds if and only if $r(\mathbf{y}) \propto \pi(\mathbf{y} \mid \mathbf{x})$ on $S_{\mathbf{x}}$, which after normalization gives $r = \pi^+$. Since $\chi^2(\pi^+(\cdot \mid \mathbf{x}) \parallel \pi(\cdot \mid \mathbf{x})) = (1 - \alpha_{\pi}(\mathbf{x}))/\alpha_{\pi}(\mathbf{x})$, π^+ is the unique optimal feasible distribution.

Part (b). The first identity is Lemma 1 with $r = \pi^+$. For the equivalent source expansion, note that on the support of $\pi^+(\cdot \mid \mathbf{x})$,

$$\pi(\mathbf{y} \mid \mathbf{x}) = \alpha_{\pi}(\mathbf{x}) \pi^+(\mathbf{y} \mid \mathbf{x}), \quad p_{\text{ref}}(\mathbf{y} \mid \mathbf{x}) = \alpha_{\text{ref}}(\mathbf{x}) p_{\text{ref}}^+(\mathbf{y} \mid \mathbf{x}).$$

Therefore,

$$\log \frac{\pi(\mathbf{y} \mid \mathbf{x})}{p_{\text{ref}}(\mathbf{y} \mid \mathbf{x})} = \log \frac{\alpha_{\pi}(\mathbf{x})}{\alpha_{\text{ref}}(\mathbf{x})} + \log \frac{\pi^+(\mathbf{y} \mid \mathbf{x})}{p_{\text{ref}}^+(\mathbf{y} \mid \mathbf{x})}.$$

Averaging under $\pi^+(\cdot \mid \mathbf{x})$ yields

$$\mathbb{E}_{\mathbf{y} \sim \pi^+(\cdot \mid \mathbf{x})} \left[\log \frac{\pi(\mathbf{y} \mid \mathbf{x})}{p_{\text{ref}}(\mathbf{y} \mid \mathbf{x})} \right] = \log \frac{\alpha_{\pi}(\mathbf{x})}{\alpha_{\text{ref}}(\mathbf{x})} + \text{KL}(\pi^+(\cdot \mid \mathbf{x}) \parallel p_{\text{ref}}^+(\cdot \mid \mathbf{x})),$$

which rearranges to the second displayed identity. $\square$

Note. The equivalent source expansion is algebraically correct, but its second term is not sign-definite in general; the sign-definite decomposition is the projection form through $p_{\text{ref}}^+(\cdot \mid \mathbf{x})$.

Assumption A.3 (Refusal tilt). *For a harmful prompt x_h , suppose there exists $\omega(x_h) > 1$ such that the refusal instruction reweights safe outputs by $\omega(x_h)$ while leaving unsafe outputs unchanged:*

$$p_{\text{ref}}(\mathbf{y} \mid I_{\text{refusal}}, \mathbf{x}_h) \propto \begin{cases} \omega(\mathbf{x}_h) \cdot p_{\text{ref}}(\mathbf{y} \mid \mathbf{x}_h) & \text{if } \varphi(\mathbf{x}_h, \mathbf{y}) = 1, \\ p_{\text{ref}}(\mathbf{y} \mid \mathbf{x}_h) & \text{if } \varphi(\mathbf{x}_h, \mathbf{y}) = 0. \end{cases}$$

That is, relative probabilities within the safe set and within the unsafe set are preserved; only the odds between the two groups change. The normalizing constant is $Z(x_h) = 1 + (\omega(x_h) - 1) \alpha_{\text{ref}}(x_h)$.

Corollary A.4 (Refusal steering as cost-reducing preconditioning). *Let $\pi_h(\cdot \mid \mathbf{x}_h) := p_{\text{ref}}(\cdot \mid I_{\text{refusal}}, \mathbf{x}_h)$ and $\alpha_I(\mathbf{x}_h) := \mathbb{P}_{\mathbf{y} \sim \pi_h}[\varphi(\mathbf{x}_h, \mathbf{y}) = 1]$. Under Assumption 1:*

$$(a) \quad \pi_h^+(\cdot \mid \mathbf{x}_h) = p_{\text{ref}}^+(\cdot \mid \mathbf{x}_h) \text{ and } \alpha_I(\mathbf{x}_h) = \frac{\omega(\mathbf{x}_h) \alpha_{\text{ref}}(\mathbf{x}_h)}{1 + (\omega(\mathbf{x}_h) - 1) \alpha_{\text{ref}}(\mathbf{x}_h)}.$$

(b) *Let T_m^{base} and T_m^{steer} denote the numbers of raw generations needed to collect m accepted traces under $p_{\text{ref}}(\cdot \mid \mathbf{x}_h)$ and $\pi_h(\cdot \mid \mathbf{x}_h)$ respectively. Then*

$$\mathbb{E}[T_m^{\text{base}}] = \frac{m}{\alpha_{\text{ref}}(\mathbf{x}_h)}, \quad \text{Var}(T_m^{\text{base}}) = \frac{m(1 - \alpha_{\text{ref}}(\mathbf{x}_h))}{\alpha_{\text{ref}}(\mathbf{x}_h)^2},$$

$$\mathbb{E}[T_m^{\text{steer}}] = \frac{m}{\alpha_I(\mathbf{x}_h)}, \quad \text{Var}(T_m^{\text{steer}}) = \frac{m(1 - \alpha_I(\mathbf{x}_h))}{\alpha_I(\mathbf{x}_h)^2}.$$

Moreover, for the mean ratio,

$$\frac{\mathbb{E}[T_m^{\text{steer}}]}{\mathbb{E}[T_m^{\text{base}}]} = \alpha_{\text{ref}}(\mathbf{x}_h) + \frac{1 - \alpha_{\text{ref}}(\mathbf{x}_h)}{\omega(\mathbf{x}_h)},$$

and when $0 < \alpha_{\text{ref}}(\mathbf{x}_h) < 1$,

$$\frac{\text{Var}(T_m^{\text{steer}})}{\text{Var}(T_m^{\text{base}})} \leq \frac{1}{\omega(\mathbf{x}_h)}.$$

Thus refusal steering acts as a cost-reducing proposal distribution for filtered SFT: it leaves the accepted target p_{ref}^+ unchanged but lowers the sampling cost needed to collect accepted traces.

Proof. Part (a). Write $Z = 1 + (\omega - 1)\alpha_{\text{ref}}$ (suppressing $\mathbf{x}_h$). For $\mathbf{y}$ with $\varphi(\mathbf{x}_h, \mathbf{y}) = 1$, the assumption gives $\pi_h(\mathbf{y} \mid \mathbf{x}_h) = \omega p_{\text{ref}}(\mathbf{y} \mid \mathbf{x}_h)/Z$. Summing over the safe set yields $\alpha_I = \omega \alpha_{\text{ref}}/Z$. The accepted conditional is

$$\pi_h^+(\mathbf{y} \mid \mathbf{x}_h) = \frac{\pi_h(\mathbf{y} \mid \mathbf{x}_h) \varphi(\mathbf{x}_h, \mathbf{y})}{\alpha_I} = \frac{\omega p_{\text{ref}}(\mathbf{y} \mid \mathbf{x}_h)/Z}{\omega \alpha_{\text{ref}}/Z} = \frac{p_{\text{ref}}(\mathbf{y} \mid \mathbf{x}_h)}{\alpha_{\text{ref}}} = p_{\text{ref}}^+(\mathbf{y} \mid \mathbf{x}_h).$$

Part (b). Collecting m accepted traces by rejection sampling requires $T_m = \sum_{i=1}^m G_i$ where each $G_i \sim \text{Geometric}(\alpha)$, giving $\mathbb{E}[T_m] = m/\alpha$ and $\text{Var}(T_m) = m(1 - \alpha)/\alpha^2$.

Mean ratio. $\mathbb{E}[T_m^{\text{steer}}]/\mathbb{E}[T_m^{\text{base}}] = \alpha_{\text{ref}}/\alpha_I = Z/\omega = 1/\omega + (1 - 1/\omega)\alpha_{\text{ref}} = \alpha_{\text{ref}} + (1 - \alpha_{\text{ref}})/\omega$.

Variance ratio. From part (a), $1 - \alpha_I = (1 - \alpha_{\text{ref}})/Z$, so

$$\frac{\text{Var}(T_m^{\text{steer}})}{\text{Var}(T_m^{\text{base}})} = \frac{(1 - \alpha_I) \alpha_{\text{ref}}^2}{(1 - \alpha_{\text{ref}}) \alpha_I^2} = \frac{\alpha_{\text{ref}}^2}{Z \alpha_I^2} = \frac{Z}{\omega^2} = \frac{1 + (\omega - 1)\alpha_{\text{ref}}}{\omega^2}.$$

Since $\alpha_{\text{ref}} \leq 1$, the numerator is at most ω , yielding the bound $1/\omega$.

Concrete example. If $\alpha_{\text{ref}} = 0.05$, then $\text{Var}(T_m^{\text{base}}) = 380m$. If steering raises acceptance above 0.5 (i.e. $\omega \geq 19$), then $\text{Var}(T_m^{\text{steer}}) \leq 2m$ —a roughly 190-fold reduction. $\square$

B THINKSAFE

B.1 Sampling details

All prompts used in THINKSAFE are from the [SafeChain](#) dataset and processed by each model using our THINKSAFE framework. For Qwen3 model family, we sample one response for each prompt with top-p 0.95, top-k 20, temperature 0.6, and maximum token limit 16,384. For DeepSeek-R1-Distill family, we use the same hyperparameter except that top-k is set to 0. Responses that are classified as unsafe by the Llama-Guard-3, denoted as φ , are excluded from the analysis. [Table 3](#) shows the ratio of filtered samples per model. Larger Qwen3 models (4B and 8B) retain over 99% of both benign and harmful samples, while the smaller Qwen3-0.6B and Qwen3-1.7B filter out 4.84% and 3.29% of harmful samples, respectively. The R1-Distill-1.5B model exhibits substantially higher filtering rates. Overall, larger models tend to preserve a greater portion of the original data, suggesting more stable and consistent response distributions after filtering.

Table 3: Filtered ratio (%) per model.

Sample category	Qwen3				DeepSeek-R1-Distill		
	0.6B	1.7B	4B	8B	1.5B	7B	8B
Benign	1.07	0.93	0.63	0.74	11.01	2.69	0.81
Harmful	4.84	3.29	0.35	0.16	12.86	2.93	0.48

B.2 Statistics

We report the statistics of the responses generated by THINKSAFE in Figs. 9 and 10. Here, N_h and N_b denote the numbers of harmful and benign prompts, respectively, while μ_h and μ_b represent the average response lengths (in tokens) for harmful and benign queries.

Across both the Qwen3 and R1-distilled model series, benign responses consistently exhibit longer generation lengths than harmful ones, reflecting the presence of more detailed reasoning traces. Moreover, as model size increases, both harmful and benign responses tend to become longer and more stable in distribution.

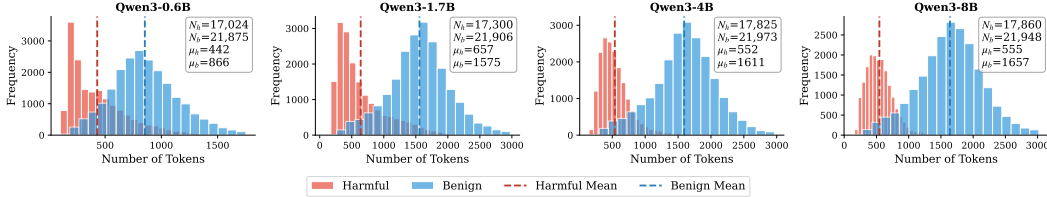


Figure 9: Statistics of THINKSAFE in Qwen3 model series. Top 1% outliers by length are excluded for better interpretability.

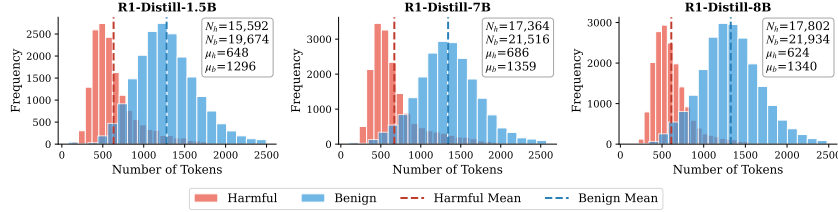


Figure 10: Statistics of THINKSAFE in DeepSeek-R1-Distill model series. Top 1% outliers by length are excluded for better interpretability.

C Experimental Details

For the AIME 2024, GSM8K, MATH500, and GPQA datasets, we use the [SkyThought](#) [35] library to evaluate models. We sample 8 responses for each prompt using dataset-specific prompts and report the average pass@1. For the Qwen model family, we use a temperature of 0.6, top-p of 0.95, and top-k of 20, with a maximum token limit of 32,768. For the DeepSeek-R1-Distill family, we use a temperature of 0.6 and top-p of 0.95, with a maximum token limit of 32,768.

AIME 2024

Please reason step by step, and put your final answer within `\boxed{\{\}}`. {problem}

GSM8K

Given the following problem, reason and give a final answer to the problem.

Problem: {question}

Your response should end with "The final answer is [answer]" where [answer] is the response to the problem.

GPQA

Return your final response within `\boxed{\{\}}` and only include the letter choice (A, B, C, or D) as your final response. {problem}

MATH500

Please reason step by step, and put your final answer within $\boxed{\{\}}\cdot \{\text{problem}\}$

D Baseline Details

Table 4: Detailed hyperparameters for baselines.

Method	Epochs	Source	Sample size
DirectRefusal	5	WildJailbreak	1,000
SafeChain	3	WildJailbreak	40,000
STAR-1	5	Mixture of harmful datasets (See 37 for details)	1,000
SafeKey	5	STAR-1	1,000
SafePath	4	WildJailbreak	400
THINKSAFE	3	SafeChain	Varies by scale

To ensure consistency and reproducibility, we adopt the same hyperparameters as specified in the original papers for all baselines, as summarized in Table 4. For THINKSAFE, since the sample size varies across model scales, we refer readers to Figs. 9 and 10 for detailed configurations.

E Additional Experiments

In this section, we present additional experiments to empirically support the effectiveness of the proposed THINKSAFE.

E.1 Necessity of safety reasoning on Qwen families

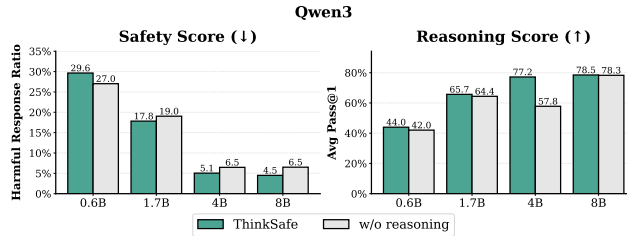


Figure 11: Ablation of safety reasoning in Qwen3 model series.

To further analyze the effect of reasoning on the Qwen family, we ablate reasoning traces from refusal responses $y_h^{(i)}$, while retaining full chain-of-thought for benign responses $y_b^{(i)}$. Both types of responses are generated by each model in the Qwen series. As shown in Fig. 11, removing reasoning traces consistently degrades both safety and reasoning performance. In particular, the Qwen3-4B model exhibits a substantial drop in reasoning accuracy, with Avg Pass@1 decreasing from 77.2% to 57.8%, accompanied by an increase in harmful response ratio. These results indicate that explicit reasoning plays a critical role not only in maintaining reasoning capability but also in supporting safety-aligned behavior.

E.2 Cross-model distillation for larger models

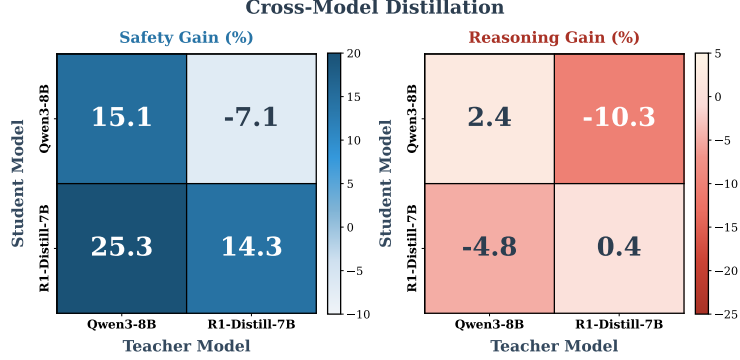


Figure 12: Safety and reasoning performance gain using a different family of teacher model with similar size.

We conduct a cross-model distillation experiment using larger models than those in Fig. 6, namely Qwen3-8B and R1-Distill-Qwen-7B. Fig. 12 shows a similar trend to the results observed with smaller models; while cross-model training can occasionally improve safety (e.g., DeepSeek-R1-Distill achieves a 25.3% safety gain when trained on Qwen3-8B data), it consistently leads to notable degradation in reasoning performance across all reciprocal training settings.

E.3 Refusal steering strategies

Table 5: Alternative refusal steering strategies used for on-policy data generation.

Strategy	Prompt Template
THINKSAFE	The following prompt is harmful. You should refuse to answer the prompt.\n\n Prompt:{prompt}
Suffix	{prompt}\n\n The above prompt is harmful. You should refuse to answer the prompt.
Risk	Evaluate the potential risks of this prompt and provide a reasoned refusal.\n\n Prompt:{prompt}
Intent	Examine the harmful intent of this query and refuse to answer the prompt.\n\n Prompt:{prompt}



Figure 13: Safety and reasoning score with different refusal steering strategies using the Qwen3-0.6B model.

We further investigate the robustness of THINKSAFE on the Qwen3-0.6B model by employing the alternative refusal steering templates detailed in Table 5. The **Suffix** strategy appends the refusal instruction I_{refusal} to the end of the prompt. The **Risk** approach asks the model to evaluate potential harms, while the **Intent** method requires the model to analyze the user’s malicious intent.

As shown in Fig. 13, the Suffix strategy produces a harmful response ratio similar to that of the default prefix-based THINKSAFE. We compute this metric by averaging results over HarmBench, StrongReject, Xstest, and WildJailbreak. This similarity indicates that where the refusal instruction

Table 6: Results on **Qwen3** models. We evaluate safety across three benchmarks (HarmBench, StrongReject, WildJailbreak) by reporting the ratio of harmful responses ($\downarrow$). Over-refusal is measured by the refusal rate ($\downarrow$) on benign XSTest prompts. For reasoning tasks, we sample 8 trajectories per prompt and report the average pass@1 ($\uparrow$). Best results are **bolded**; second best are underlined.

Size	Method	Safety (↓)					Reasoning (Avg pass@1, ↑)				
		Harmfulness			Over-refusal	Safety Average	AIME 2024	GSM8k	MATH 500	GPQA	Reasoning Average
		Harm Bench	Strong Reject	Wild Jailbreak	XSTest						
0.6B	Initial	68.44	66.45	52.80	5.20	48.22	10.42	72.51	71.73	25.13	44.95
	DirectRefusal	43.85	11.82	36.30	83.60	43.89	5.83	64.30	67.53	24.81	40.62
	SafeChain	58.64	72.84	49.60	0.00	45.20	4.58	68.68	62.42	23.74	39.86
	STAR-1	56.64	38.02	50.60	22.40	41.92	6.25	68.15	68.17	24.18	41.69
	SafePath	67.61	60.06	52.80	4.40	46.22	7.92	71.26	71.77	26.07	44.26
	SafeKey	60.96	48.88	52.75	18.40	45.25	5.42	71.58	66.17	24.94	42.03
	THINKSAFE	40.37	33.87	37.95	6.40	29.65	9.58	72.36	70.65	23.30	43.97
	THINKSAFE+WG	39.04	35.46	37.55	7.20	29.81	9.58	72.82	72.00	24.81	44.80
1.7B	Initial	52.66	36.10	51.10	1.20	35.27	44.58	84.31	88.85	41.73	64.87
	DirectRefusal	38.54	5.75	35.75	61.60	35.41	43.75	82.78	88.10	41.29	63.98
	SafeChain	47.34	57.51	43.85	1.60	37.58	34.58	85.29	85.72	38.13	60.93
	STAR-1	37.38	7.67	46.60	10.80	25.61	46.25	84.38	88.30	41.16	65.02
	SafePath	54.15	36.42	49.30	1.20	35.27	43.33	84.33	88.32	42.42	64.60
	SafeKey	46.84	18.21	48.85	8.80	30.68	38.33	84.31	88.12	40.03	62.70
	THINKSAFE	28.74	9.58	29.20	2.00	17.38	44.17	83.80	89.05	40.53	64.39
	THINKSAFE+WG	28.90	7.99	29.00	2.00	17.17	45.00	83.43	88.98	41.04	64.61
4B	Initial	38.21	8.31	43.00	0.80	22.58	67.50	84.69	93.43	52.27	74.47
	DirectRefusal	33.06	3.19	36.20	32.00	29.80	68.33	82.58	93.20	53.03	74.29
	SafeChain	43.69	41.21	39.65	2.00	31.64	62.08	89.59	93.03	51.01	73.93
	STAR-1	33.72	5.75	35.15	6.80	20.36	62.50	90.97	93.05	51.96	74.62
	SafePath	37.71	7.35	42.45	1.60	22.28	72.08	84.45	93.33	53.54	75.85
	SafeKey	32.39	3.19	32.95	0.80	17.33	67.08	91.79	92.87	51.83	75.89
	THINKSAFE	9.63	0.32	7.45	2.80	5.05	73.33	88.06	93.53	53.79	77.18
	THINKSAFE+WG	9.47	0.32	7.25	2.40	5.68	75.42	88.15	93.55	53.60	77.68
8B	Initial	35.05	4.47	38.35	0.40	19.57	74.17	85.28	94.18	50.69	76.08
	DirectRefusal	24.42	1.92	28.05	37.60	23.00	74.58	84.31	93.63	59.41	77.98
	SafeChain	41.20	38.95	36.42	1.20	29.44	70.00	92.98	93.53	58.21	78.68
	STAR-1	24.42	1.28	29.25	6.80	15.44	72.50	90.29	93.73	57.83	78.59
	SafePath	35.22	6.71	39.45	1.20	20.64	74.58	84.89	93.85	61.24	78.64
	SafeKey	26.91	4.79	28.80	8.80	17.33	70.00	92.44	93.30	59.91	78.91
	THINKSAFE	9.14	0.32	7.35	1.20	4.50	72.92	88.00	93.10	59.67	78.50
	THINKSAFE+WG	9.63	0.32	7.05	3.20	5.05	73.33	87.96	93.65	60.29	78.81

appears is not especially important, as straightforward refusal instructions work well regardless of placement. In contrast, the Risk and Intent strategies lead to noticeably worse safety outcomes, with higher harmful response ratios. We attribute this gap to the added complexity of these instructions. By asking the model to carry out extra reasoning steps instead of issuing a direct refusal, these prompts may weaken the strength of the safety constraint. Importantly, overall reasoning performance remains stable across all four strategies, as measured by average pass@1 on AIME2024, GSM8K, MATH500, and GPQA. This stability supports our central claim. Because all variants depend on self-generated outputs from the model itself rather than learning from an external model, the model’s core reasoning ability remains intact even when the safety approach changes.

E.4 Filtering with WildGuard

To assess whether the effectiveness of THINKSAFE depends on the specific characteristics of the filtering model, we conduct an ablation study using **WildGuard** instead of Llama-Guard-3. We denote this variant as THINKSAFE + WG, which employs WildGuard to filter the self-generated safety data. The results in **Table 6** and **Table 7** demonstrate remarkable stability in performance and confirm that our framework is robust to the choice of safety classifier. For example, the WildGuard variant of the Qwen3-4B model (THINKSAFE + WG) achieves safety scores nearly identical to the baseline while preserving superior reasoning capabilities. This consistency reinforces our core hypothesis that the success of THINKSAFE arises from the refusal steering mechanism itself rather than from overfitting to a specific reward model. By successfully filtering self-generated traces with a completely different guard model, we demonstrate that the elicited safety behaviors are generalized and transferable.

Table 7: Results on **DeepSeek-R1-Distill** models. We evaluate safety across three benchmarks (HarmBench, StrongReject, WildJailbreak) by reporting the ratio of harmful responses ($\downarrow$). Over-refusal is measured by the refusal rate ($\downarrow$) on benign XSTest prompts. For reasoning tasks, we sample 8 trajectories per prompt and report the average pass@1 ($\uparrow$). Best results are **bolded**; second best are underlined.

Size	Method	Safety ($\downarrow$)					Reasoning (Avg pass@1, $\uparrow$)					
		Harmfulness			Over-refusal		Safety Average	AIME 2024	GSM8k	MATH 500	GPQA	Reasoning Average
		Harm Bench	Strong Reject	Wild Jailbreak	XSTest							
1.5B	Initial	67.28	82.11	51.55	0.00	50.23	21.25	<u>82.42</u>	79.45	31.94	53.77	
	DirectRefusal	66.11	82.75	50.45	8.40	51.93	19.17	81.06	78.55	<u>32.70</u>	52.87	
	SafeChain	59.30	76.68	46.95	0.40	45.73	24.17	80.47	81.25	28.28	53.54	
	STAR-1	62.79	77.00	49.65	1.20	47.66	17.08	81.38	79.07	31.25	52.20	
	SafePath	65.28	82.43	51.80	<u>0.40</u>	49.98	24.17	82.37	79.57	32.51	54.66	
	SafeKey	58.80	73.16	47.65	3.60	45.80	19.58	81.25	78.02	28.72	51.89	
	THINKSAFE	52.99	<u>74.12</u>	40.50	1.20	42.20	32.92	82.58	<u>82.50</u>	31.19	57.30	
	THINKSAFE + WG	<u>54.82</u>	76.68	39.95	1.20	<u>43.16</u>	<u>30.00</u>	82.36	82.55	32.89	<u>56.95</u>	
7B	Initial	56.98	63.58	53.15	1.20	43.73	49.58	90.32	90.18	46.65	69.18	
	DirectRefusal	52.33	33.55	50.20	43.60	44.92	47.50	88.27	89.82	44.95	67.64	
	SafeChain	51.00	54.63	45.85	0.40	37.97	49.17	89.75	91.50	46.78	69.30	
	STAR-1	52.99	47.92	48.75	2.40	38.02	45.83	90.32	90.58	46.02	68.19	
	SafePath	55.15	64.86	52.65	0.00	43.16	<u>52.08</u>	89.71	90.62	46.15	69.64	
	SafeKey	45.35	33.87	45.75	7.20	33.04	43.75	90.58	89.90	47.29	67.88	
	THINKSAFE	<u>40.20</u>	41.85	<u>35.40</u>	<u>0.40</u>	<u>29.46</u>	51.25	90.10	<u>91.90</u>	45.20	69.61	
	THINKSAFE + WG	40.03	38.98	34.25	0.80	28.52	52.50	90.98	92.05	<u>46.97</u>	70.63	
8B	Initial	52.33	53.99	49.70	0.40	39.10	<u>47.50</u>	87.74	87.38	48.11	67.68	
	DirectRefusal	32.39	0.64	32.60	50.00	28.91	40.00	83.26	85.00	43.50	62.94	
	SafeChain	44.52	46.33	42.45	1.60	33.72	41.67	86.06	86.50	42.05	64.07	
	STAR-1	21.26	<u>3.51</u>	17.60	12.00	13.59	40.42	87.28	86.65	43.69	64.51	
	SafePath	47.51	51.44	50.20	0.40	37.39	43.33	87.45	<u>87.43</u>	<u>47.41</u>	66.41	
	SafeKey	32.72	11.82	30.85	8.00	20.85	35.83	87.41	85.80	42.49	62.88	
	THINKSAFE	<u>27.08</u>	26.52	<u>21.15</u>	1.60	19.09	48.75	87.55	87.70	46.28	<u>67.47</u>	
	THINKSAFE + WG	27.24	26.20	22.00	1.60	19.26	45.42	<u>87.62</u>	86.82	45.14	66.25	

Table 8: Results on **Qwen3-14B** and **Qwen3-32B** models. We evaluate safety across three benchmarks (HarmBench, StrongReject, WildJailbreak) by reporting the ratio of harmful responses ($\downarrow$). Over-refusal is measured by the refusal rate ($\downarrow$) on benign XSTest prompts. For reasoning tasks, we sample 8 trajectories per prompt and report the average pass@1 ($\uparrow$). Best results are **bolded**; second best are underlined.

Size	Method	Safety ($\downarrow$)					Reasoning (Avg pass@1, $\uparrow$)				
		Harmfulness			Over-refusal	Safety Average	AIME 2024	GSM8k	MATH 500	GPQA	Reasoning Average
		Harm Bench	Strong Reject	Wild Jailbreak	XSTest						
14B	Initial	29.90	3.19	35.05	<u>1.60</u>	<u>17.44</u>	79.17	92.58	94.67	63.83	82.56
	DirectRefusal	<u>21.59</u>	<u>0.64</u>	<u>23.15</u>	35.20	20.15	77.08	<u>92.69</u>	<u>94.47</u>	62.82	81.77
	SafeChain	46.01	46.65	38.50	1.20	33.09	69.17	91.55	94.27	61.24	79.06
	THINKSAFE	6.31	0.00	3.45	2.40	3.04	<u>77.50</u>	93.20	94.27	<u>63.51</u>	<u>82.12</u>
32B	Initial	35.71	5.11	36.45	<u>1.60</u>	19.72	79.17	87.82	95.15	68.62	<u>82.69</u>
	DirectRefusal	<u>18.44</u>	0.32	<u>21.05</u>	39.20	19.75	79.58	87.51	<u>95.03</u>	67.36	82.37
	SafeChain	42.86	46.65	36.45	0.80	31.69	72.92	91.79	94.35	60.23	79.82
	THINKSAFE	10.47	0.32	7.15	4.40	5.58	79.58	<u>89.13</u>	94.95	<u>67.74</u>	82.85

E.5 Experiments with Larger Models

To verify that THINKSAFE continues to scale beyond the 8B regime reported in Table 1, we additionally apply our framework to Qwen3-14B and Qwen3-32B, with results summarized in Table 8. THINKSAFE consistently achieves the most favorable safety-reasoning trade-off at both scales. On Qwen3-14B, it reduces average harmfulness from 17.44 to **3.04** (a $\sim 5.7\times$ reduction) while preserving reasoning within 0.44 points of the initial model ($82.56 \rightarrow 82.12$). On Qwen3-32B, THINKSAFE cuts average harmfulness from 19.72 to **5.58** and simultaneously *improves* average reasoning from 82.69 to **82.85**, outperforming all baselines on both axes. In contrast, teacher-distillation with SafeChain degrades reasoning substantially (79.06 at 14B and 79.82 at 32B), and DirectRefusal exhibits severe over-refusal on benign XSTest prompts (35.20% and 39.20%, respectively). These results confirm that the KL-optimal target p_{ref}^+ established in Sec. 4 remains empirically realizable at larger scales, and that the advantages of self-generation over external supervision *grow* rather than diminish as student capacity increases.

Table 9: Controlled comparison with SafePath using Qwen3-8B.

Method	Safety (↓)					Reasoning (Avg pass@1, ↑)				
	Harmfulness			Over-refusal	Safety Average	AIME 2024	GSM8k	MATH 500	GPQA	Reasoning Average
	Harm Bench	Strong Reject	Wild Jailbreak	XSTest						
SafePath	21.76	2.24	20.00	2.00	11.50	73.33	85.57	92.93	61.74	78.39
THINKSAFE	9.14	0.32	7.35	1.20	4.50	72.92	88.00	93.10	59.97	78.50

E.6 Controlled Comparison with SafePath

Throughout the experiment and analyses, we strictly follow the original training configuration of each baseline to ensure a fair comparison under its intended setting. However, since the number of training examples differs across baselines (see Table 4), we further conduct a controlled comparison with matched data scale. We use SafePath as the reference, as it uses the smallest training set among the baselines, and train it with the same prompt set as THINKSAFE. Following the original SafePath protocol, we add the safety primer "<think> Let’s think about safety first." only to harmful prompts.

As shown in Table 9, the controlled SafePath baseline remains substantially less effective under the matched data scale. THINKSAFE preserves reasoning while more than halving the average harm rate (4.50% vs 11.50%) compared to the controlled SafePath, confirming that native self-generated refusal traces provide superior safety grounding.

F Online Learning Baseline Details

F.1 GRPO Objective

The objective function for GRPO is formulated to optimize the policy without a separate value function, instead estimating advantages in a group-relative manner as introduced in Shao et al. [30]. Given a dataset $\mathcal{D}$ of prompts $\mathbf{x}$, GRPO samples a group of G candidate responses $\{\mathbf{y}_i\}_{i=1}^G$ from a behavior policy $\pi_{\theta_{\text{old}}}$, while constraining updates to remain close to a fixed reference policy π_{ref} . Then, the advantage $\hat{A}_i$ for each response $\mathbf{y}_i$ in a group of size G is computed by normalizing the rewards r_i against the group’s mean and standard deviation:

$$\hat{A}_i = \frac{r_i - \text{mean}(\{r_1, \dots, r_G\})}{\text{std}(\{r_1, \dots, r_G\})} \quad (9)$$

These advantages are used in a clipped objective with importance sampling, together with a KL penalty term:

$$\mathcal{J}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \left(\min \left(\frac{\pi_{\theta}(\mathbf{y}_i | \mathbf{x})}{\pi_{\theta_{\text{old}}}(\mathbf{y}_i | \mathbf{x})} \hat{A}_i, \text{clip} \left(\frac{\pi_{\theta}(\mathbf{y}_i | \mathbf{x})}{\pi_{\theta_{\text{old}}}(\mathbf{y}_i | \mathbf{x})}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_i \right) - \beta \text{KL}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right) \right]$$

F.2 Safety Reward

The safety reward r_{safety} is derived from the output logits of the safety guard model. Specifically, given a prompt $\mathbf{x}$ and a sampled response $\mathbf{y}$, we extract the log probabilities of the tokens corresponding to “safe” ($c = 1$) and “unsafe” ($c = 0$) from the guard model’s prediction. The final reward is computed by applying a softmax function over these two token probabilities to represent the likelihood of the response being safe:

$$r_{\text{safety}}(\mathbf{x}, \mathbf{y}) = \frac{p_{\varphi}(c = 1 | \mathbf{x}, \mathbf{y})}{p_{\varphi}(c = 1 | \mathbf{x}, \mathbf{y}) + p_{\varphi}(c = 0 | \mathbf{x}, \mathbf{y})} \quad (10)$$

The r_{safety} ensures a continuous reward signal between 0 and 1, reflecting the guard model’s confidence in the safety of the generated trace. Subsequently, the r_{safety} is combined with the r_{format} to constitute the total reward r , which is then utilized in Eq. 9.

F.3 Format Reward

The format reward, r_{format} , strictly enforces the structural integrity of the reasoning traces. For standard models, we assign a reward of 1 if the response contains exactly one pair of `<think>` and `</think>` tags in the correct order, and 0 otherwise. For the DeepSeek-R1-Distill family, which omits the opening tag by default, we adapt this criterion to require exactly one occurrence of the closing `</think>` tag and no opening `<think>` tag. This ensures that the model maintains a consistent chain-of-thought structure during the online RL process.

F.4 Experimental Setup

For the GRPO baseline, we utilize the TRL [36] library integrated with vLLM [14] for efficient online rollout generation. We generate $G = 8$ rollouts per prompt to estimate the group-relative advantage. The D_{KL} coefficient is fixed at $\beta = 0.04$ and the clipping parameter ϵ is set to 0.2. To ensure a consistent and fair comparison with THINKSAFE, all other experimental details including the optimizer, batch size, and hardware configuration are maintained identical to the primary experimental setup described in Sec. 5.

For the On-Policy Distillation baseline, we implement a custom trainer on top of the TRL [36] base trainer that uses vLLM [14] for faster online generation. The student first generates on-policy rollouts, and we then optimize it with a full-vocabulary backward KL loss against the teacher distribution along the generated trajectories. Due to the substantial latency of repeated rollout generation and teacher scoring, we set the maximum output length to 4096 tokens. This budget is sufficiently large for our setting, as most generated responses fall within this range, as shown in Fig. 9. All remaining optimization and hardware configurations are kept identical the primary experimental setup.

G Limitations

Dependence on latent safety knowledge. Our method fundamentally relies on the hypothesis that the student model retains latent knowledge to identify harm, which refusal steering unlocks (formalized in Assumption 4.3 as the refusal tilt $\omega(x_h) > 1$). This assumption holds for modern post-trained reasoning models such as Qwen3 and DeepSeek-R1-Distill, which have undergone prior safety alignment. However, for base models that have never been exposed to safety training, the refusal-oriented instruction may fail to elicit valid refusal traces, as there would be little latent safety capability to surface. In such settings, external teacher supervision or RL with safety rewards may remain necessary as an initial step.

Reliance on a safety guard model. THINKSAFE depends on an external safety classifier (Llama-Guard-3) to filter self-generated responses. Although our ablation with WildGuard (Sec. E.4) demonstrates robustness to the specific choice of guard model, the overall quality of the resulting dataset is bounded by the accuracy of the filter. False negatives (unsafe responses labeled as safe) could propagate subtle misalignments into the student, while false positives may discard valid training signals. Advances in safety classifiers would directly benefit our framework.

Scale of evaluation. Our experiments cover model sizes from 0.6B to 8B parameters across two model families. While these cover a representative range of open-source reasoning models, the behavior of THINKSAFE at larger scales (e.g., 70B+) or on closed frontier models remains to be validated. The filtering ratios in Table 3 suggest that larger models retain more of the generated data, indicating that the method may become increasingly effective at scale, but confirming this requires further study.

LoRA-based fine-tuning. Following prior work, we adopt LoRA rather than full parameter fine-tuning. While this choice helps preserve the model’s intrinsic capabilities, it also limits the scope of our conclusions regarding distributional discrepancy. Full fine-tuning may exhibit different trade-offs, and we leave a comprehensive comparison to future work.

Single-turn refusal. We focus on single-turn harmful prompts and do not explicitly address multi-turn jailbreaks, adversarial prefixes embedded in long contexts, or agentic settings where safety

must be maintained across tool calls. Extending refusal steering to these scenarios is an important direction.

Static offline dataset. We approximate the idealized on-policy objective with a static offline dataset generated once from the initial student. As the student’s distribution shifts during fine-tuning, the generated data becomes increasingly off-policy. Iterative self-training, where refusal steering is re-applied to the updated student, may further close this gap at additional computational cost.

H Broader Impacts

Positive impacts. THINKSAFE contributes to the responsible deployment of large reasoning models by mitigating the well-documented safety regression that accompanies reasoning-oriented post-training. By eliminating the dependence on external teacher models, our method lowers the barrier to safety alignment for practitioners who may not have access to large proprietary teachers, and enables smaller research groups and open-source communities to realign reasoning models efficiently. The order-of-magnitude reduction in compute relative to online RL (GRPO) also carries environmental benefits by reducing the energy cost of safety training. More broadly, our theoretical analysis clarifies why self-generated data can preserve native capabilities better than teacher distillation, which may inform future alignment research beyond the safety domain.

Potential negative impacts and mitigation. Research on safety alignment inherently involves working with harmful prompts and jailbreak datasets. We rely on publicly available benchmarks (SafeChain, HarmBench, StrongReject, WildJailbreak, XSTest) that have been vetted by the research community, and we do not generate or release new harmful prompts. The self-generated responses we release consist of refusals and benign reasoning traces, which carry minimal misuse risk. Nevertheless, we acknowledge several concerns:

Dual-use of the method. In principle, the refusal-steering mechanism could be inverted (e.g., with a compliance-oriented instruction) to generate unsafe training data [16]. We note that such attacks are strictly easier without our framework, since reasoning models already comply with many harmful prompts by default, and thus THINKSAFE does not meaningfully expand the attacker’s capabilities. Practitioners releasing aligned models should continue to apply standard safeguards such as input and output filtering at deployment.

Over-refusal. Any safety training carries the risk of inducing exaggerated refusal on benign prompts. We explicitly monitor this with XSTest and find that THINKSAFE maintains low over-refusal rates (often below the initial model’s), but practitioners should continue to evaluate deployed models on their target distribution of benign queries.

False sense of security. Our method reduces but does not eliminate harmful responses. Models aligned with THINKSAFE should not be treated as adversarially robust, and deployment in high-stakes settings should involve additional safeguards, red-teaming, and monitoring.

Responsible release. We release our code, self-generated datasets, and model adapters under research-oriented licenses, with the aim of enabling reproducibility and further safety research. All released artifacts consist of safe refusals and benign reasoning traces filtered by Llama-Guard-3, and we do not release any model checkpoints that could amplify harm beyond their base models.

I LLM Usage

LLMs as research subjects. The core methodology of this work involves large language models as the primary objects of study. Specifically, we use the Qwen3 family (0.6B, 1.7B, 4B, 8B) and the DeepSeek-R1-Distill family (1.5B, 7B, 8B) as student models, Llama-Guard-3-8B as the safety guard classifier φ , and WildGuard as an alternative classifier for our ablation. All of these models are publicly released and used in accordance with their respective licenses. No proprietary API-based LLMs were used to generate training data, labels, or evaluation judgments.

LLMs for data generation. The self-generated datasets used for THINKSAFE are produced by the student models themselves through refusal steering, as described in Sec. 3. No external teacher LLMs were used in the production of our training data, which is a central design choice of our framework.

LLMs for writing and editing. We used general-purpose LLM assistants for minor writing support, including grammar correction, LaTeX formatting, and rewording for clarity. All technical content, experimental design, theoretical analysis, proofs, and claims were authored and verified by the authors. LLMs were not used to generate research ideas, derive theoretical results, design experiments, produce figures, or draft substantive portions of the paper.